

As an election official, I want to be able to run a Single Transferable Vote (STV) election as requested by the Secretary of State.

Acceptance Criteria:

- Fair tie-breaker for in the case (edge case) where multiple candidates receive 0 votes.
- The STV election should produce the correct output so that future STV elections can be conducted correctly and fairly.
- Output a correct list of winners and losers in order of the election progress.
- The election summary should be shown on the screen at the end of the election, and an audit file for inspection should be generated.

Definition of Done:

- Refactor the code.
- Fix the STV code of the edge case where multiple candidates receive 0 votes. There should be a fair tie-breaker.
- Pass all unit/system tests.
- Complete the corresponding testing log and documentation.
- Able to compile and run on CSE lab machine

Effort: Small, 1-2 hours.

PBI Author: Mark Tsai, Michael Dunn.

As an election official, I want to make sure that each incoming ballot running a STV algorithm is validated as requested by the Secretary of State.

Acceptance Criteria:

- At least half of the candidates should be ranked for a ballot to be valid.
- All removed ballots must be included in the audit file.
- The total number of ballots must be adjusted, subtracting from the number of invalid ballots detected.

Definition of Done:

- Refactor the code.
- Implement the function in the STV algorithm.
- Pass all unit/system tests.
- Complete the corresponding testing log and documentation.
- Able to compile and run on CSE lab machine

Effort: Medium, 3-4 hours.

PBI Author: Manan Chaturvedi, Mark Tsai.

As an election official, I want a small report to be run after each election for all types of elections (STV, MV, Plurality) that summarizes their outcomes. This is so that all elections have their respective critical information available for referring back to validate the election and ensure an integrous election is run.

Acceptance Criteria:

- There exists a small report for each election containing valid information about the election's total number of seats and ballots, candidate names and the total number of votes/percentage of total votes, and the list of winners.

Definition of Done:

- Refactor the code.
- Implement the small report function.
- Pass all unit/system tests.
- Integrate into the system with system testing functionality.
- Complete the corresponding testing log and documentation.
- Able to compile and run on CSE lab machine

Effort: Small, 1-2 hours.

PBI Author: Manan Chaturvedi.